

11th International Workshop on Finite Elements for Microwave Engineering – FEM2012**June 4-6, 2012, Estes Park, Colorado, USA**<http://www.engr.colostate.edu/FEM2012>**General Chairs:** Branislav M. Notaros, Colorado State University, notaros@colostate.edu, and Jian-Ming Jin, University of Illinois, j-jin1@ad.uiuc.edu**Technical Program Committee Chair:** Milan M. Ilic, University of Belgrade and Colorado State University, milanilic@ieee.org**TECHNICAL AND SOCIAL PROGRAM****Overall Technical Program (Preliminary)**

The Historic Stanley Hotel		Monday June 4	Monday June 4	Tuesday June 5	Tuesday June 5	Wednesday June 6
		Billiard Room	Music Room	Billiard Room	Music Room	Billiard Room
8:00	Slot 1	Paper 1.1	Paper 2.1	Paper 7.1	Paper 8.1	
8:20	Slot 2	Paper 1.2	Paper 2.2	Paper 7.2	Paper 8.2	
8:40	Slot 3	Paper 1.3	Paper 2.3	Paper 7.3	Paper 8.3	
9:00	Slot 4	Paper 1.4	Paper 2.4	Paper 7.4	Paper 8.4	
9:20	Slot 5	Paper 1.5	Paper 2.5	Paper 7.5	Paper 8.5	
9:40	Coffee break					
10:10	Slot 6	Paper 1.6	Paper 2.6	Paper 9.1	Paper 8.5	
10:30	Slot 7	Paper 1.7	Paper 4.1	Paper 9.2	Paper 8.6	
10:50	Slot 8	Paper 3.1	Paper 4.2	Paper 9.3	Paper 8.7	
11:10	Slot 9	Paper 3.2	Paper 4.3	Paper 9.4	Paper 10.1	
11:30	Slot 10	Paper 3.3	Paper 4.4	Paper 9.5	Paper 10.2	
11:50	Lunch break					
13:10	Slot 11	Paper 3.4	Paper 4.5	Paper 9.6	Paper 10.3	
13:30	Slot 12	Paper 3.5	Paper 4.6	Paper 9.7	Paper 10.4	15:00-15:45
13:50	Slot 13	Paper 3.6	Paper 6.1	Paper 11.1	Paper 12.1	Open forum
14:10	Slot 14	Paper 5.1	Paper 6.2	Paper 11.2	Paper 12.2	discussion 1
14:30	Slot 15	Paper 5.2	Paper 6.3	Paper 11.3	Paper 12.3	15:45-16:30
14:50	Slot 16	Paper 5.3	Paper 6.4	Paper 11.4	Paper 12.4	Open forum
15:10	Coffee break					discussion 2
15:40	Slot 17	Paper 5.4	Paper 6.5	Paper 11.5	Paper 12.5	16:30-16:50
16:00	Slot 18	Paper 5.5	Paper 6.6	Paper 11.6	Paper 12.6	Coffee break
16:20	Slot 19	Paper 5.6	Paper 6.7	Paper 11.7	Paper 12.7	16:50-18:00
16:40	Slot 20	Paper 5.7	Paper 6.8	Paper 11.8	Paper 12.8	Open forum
17:00	Slot 21	Paper 5.8	Paper 6.9	Paper 11.9	Paper 12.9	discussion 3
17:20	Technical program ends for the day					

S1 Modeling and Design of Antennas and Arrays Using FEM (R. Kindt, D. Filipovic, E. Topsakal)
 S2 Discontinuous Galerkin Methods (Q. H. Liu, W. Cai)
 S3 Time-Domain FEM and Applications (D. White, O. Biro, T. Rylander)
 S4 FEM and CEM Applications in Optics and Nanophotonics (E. Simsek)
 S5 Model Order Reduction Techniques (R. Dyczij-Edlinger, V. de la Rubia)
 S6 Advances in Vector Bases for CEM (R. Graglia, A. Peterson, D. Wilton)
 S7 FETD Modeling of Complex Media and Structures (R. Lee, F. Teixeira)
 S8 Adaptive FEM, Higher Order Bases, and Advanced FEM Formulations (J. Webb, M. Gavrilovic, M. S.
 S9 Advances in Hybrid Methods and Multiphysics Problems (B. Shanker, L. Kempel)
 S10 FEM Modeling and Applications of Metamaterials and Periodic Media (K. Sertel, J. Volakis)
 S11 Domain-Decomposition Methods (J.-F. Lee, Z. Peng)
 S12 Advanced FEM/MoM Modeling, Design, and Optimization (C.J. Reddy, U. Jakobus, J. Zapata, J. G

Open forum discussion 1: Open Forum Discussion on FEM and CEM Bases, Elements, and Formulat
 (A. Peterson, D. Wilton, J. Webb, K. Sertel)
 Open forum discussion 2: Open Forum Discussion on DDM, MOR, and Efficient FEM and Hybrid Soluti
 (J.-F. Lee, Zhen Peng, R. Dyczij-Edlinger, V. de la Rubia)
 Open forum discussion 3: Open Forum Discussion on FEM-Based Modeling, Design, and Applications
 (L. Kempel, B. Shanker, R. Kindt, C.J. Reddy, K. Zhao, T. Euler)

*(Scroll down for Detailed Technical Program)***Technical and Social Program at a Glance (Preliminary)****Sunday, June 3, 2012**

Welcome Reception 19:00-21:00

Monday, June 4, 2012

Breakfast 6:30-8:00

Technical Sessions 8:00-11:50 (coffee break 9:40-10:10)

Lunch 11:50-13:10

Technical Sessions 13:10-17:20 (coffee break 15:10-15:40)

Stanley Hotel Ghost & History Tour – Monday Tour 17:30-19:00

Conference Banquet 19:30-22:30

Tuesday, June 5, 2012

Breakfast 6:30-8:00

Technical Sessions 8:00-11:50 (coffee break 9:40-10:10)

Lunch 11:50-13:10

Technical Sessions 13:10-17:20 (coffee break 15:10-15:40)

Stanley Hotel Ghost & History Tour – Tuesday Tour 17:30-19:00

Wednesday, June 6, 2012

Breakfast 6:30-8:00

Trip to Rocky Mountain National Park 8:00-14:00

Open Forum Discussions 15:00-18:00 (coffee break 16:30-16:50)

Closing Reception 18:00-20:00

Detailed Technical Program (Preliminary)**Monday, June 4, 2012, Billiard Room****Session 1: Modeling and Design of Antennas and Arrays Using FEM (R. Kindt, D. Filipovic, E. Topsakal)**

8:00 Paper 1.1. Highly Coupled Bow-Tie Array Design with Passive Elements
Safa Salman, Dimitris Psychoudakis, and John L. Volakis*

8:20 Paper 1.2. Fully Overlapping Domain Decomposition in Simulating Irregularly Arranged Arrays in Large Media
Tao Peng, Kubilay Sertel, and John L. Volakis*

8:40 Paper 1.3. Modeling of Antennas and Arrays Using Domain Decomposition Method
Kezhong Zhao and Nancy Lambert*

- 9:00 Paper 1.4. Theoretical Analysis of Varactor-loaded Half-width Leaky-wave Antenna Arrays
Leo C. Kempel, B. Shanker, Edward J. Rothwell, and Prem Chahal*
- 9:20 Paper 1.5. Non-conformal Domain Decomposition Methods for Modeling Large Finite Antenna Arrays
Zhen Peng, Kheng-Hwee Lim, and Jin-Fa Lee
- 9:40 Coffee break
- 10:10 Paper 1.6. Investigation of the Scattering Behavior of Transponder Antennas in Case of Nonlinear Termination
Thomas Bauernfeind, Kurt Preis, Gergely Koczka, and Oszkar Biro*
- 10:30 Paper 1.7. On the Design of Finite Planar Ultrawideband Modular Antenna (PUMA) Arrays using Domain Decomposition Methods
*Steve S. Holland, Jack T. Logan, Rick W. Kindt, and Marinos N. Vouvakis**

Session 3: Time-Domain FEM and Applications (D. White, O. Biro, T. Rylander)

- 10:50 Paper 3.1. Numerical Simulation of HPM Devices with ICEPIC
*Andrew D. Greenwood**
- 11:10 Paper 3.2. Time Domain Method for Periodic Nonlinear Eddy Current Problems
Oszkár Bíró, Gergely Koczka, and Kurt Preis
- 11:30 Paper 3.3. Hybrid FETD/FDTD Techniques Based on the Discontinuous Galerkin Method
Bao Zhu, Jiefu Chen, Wanxie Zhong, and Qing Huo Liu
- 11:50 Lunch break
- 13:10 Paper 3.4. Transient Thermal Analysis using a Non-conformal Domain Decomposition Approach
Yang Shao, Zhen Peng, and Jin-Fa Lee*
- 13:30 Paper 3.5. An Explicit and Unconditionally Stable Time-Domain Finite-Element Method of Linear Complexity
Qing He, Duo Chen, and Dan Jiao
- 13:50 Paper 3.6. Conformal PML Modeling in DGTD using Continuous Material Properties
Jue Wang, Zhen Peng, and Jin-Fa Lee*

Session 5: Model Order Reduction Techniques (R. Dyczij-Edlinger, V. de la Rubia)

- 14:10 Paper 5.1. Reduced-order Electromagnetic Modeling in the Presence of Uncertainty
*Juan S. Ochoa and Andreas C. Cangellaris**
- 14:30 Paper 5.2. Efficient Model Order Reduction for FEM Analysis of Waveguide Structures and Resonators
*Fotyga G. *, Kulas L., Nyka K., and Mrozowski M.*
- 14:50 Paper 5.3. Model Order Reduction for Systems with Geometrical Parameters
Kynthia K. Stavarakakis, Tilmann Wittig, Wolfgang Ackermann, and Thomas Weiland*
- 15:10 Coffee break
- 15:40 Paper 5.4. Reduced-Order Models for Geometric Parameters
Stefan Burgard, Ortwin Farle, and Romanus Dyczij-Edlinger*
- 16:00 Paper 5.5. Seamless Reduced Basis Element Method for Waveguide Simulation
Yanlai Chen, Jan S. Hesthaven, and Yvon Maday*
- 16:20 Paper 5.6. The Reduced Basis Method for Saddle Point Problems
Karen Veroy and Anna-Lena Gerner*
- 16:40 Paper 5.7. Reliable Fast Frequency Sweep of Microwave Circuits via the Reduced-Basis Method
Valentin de la Rubia
- 17:00 Paper 5.8. Reduced Order Modeling of High Magnetic Field Magnets
C. Daversin, C. Prudhomme, C. Trophime, and S. Veys*
- 17:20 Technical program ends for the day

Monday, June 4, 2012, Music Room

Session 2: Discontinuous Galerkin Methods (Q. H. Liu, W. Cai)

- 8:00 Paper 2.1. On the Physics, Accuracy, and Efficiency of the Dual-Field Domain-Decomposition Method for Time-Domain Electromagnetic Simulation
*Xiaolei Li and Jian-Ming Jin**
- 8:20 Paper 2.2. A Full Vectorial Generalized Discontinuous Galerkin Beam Propagation Method for Optical Waveguides
*Wei Cai**
- 8:40 Paper 2.3. Non-Conformal and Parallel and Interior Penalty Discontinuous Galerkin Method for the Time-Domain Maxwell's Equations
Stylianios Dosopoulos and Jin-Fa Lee
- 9:00 Paper 2.4. Discontinuous Galerkin Methods for Ideal MHD Equations
*Fengyan Li**
- 9:20 Paper 2.5. New Spectral-Prism-Element for Layered Structures implemented in an efficient implicit DG-FETD method
*Luis E. Tobon and Qing H. Liu**

9:40 Coffee break

10:10 Paper 2.6. An Efficient Discontinuous Galerkin Finite-Element Time-Domain Method and its Application to Micro-Resistivity Imager
Jiefu Chen, Carlos Harnboure, Lance Pate, Shawn Wallace, Jay Martin, Medhat Mickael, and Mac Wisler

Session 4: FEM and CEM Applications in Optics and Nanophotonics (E. Simsek)

10:30 Paper 4.1. Electromagnetic Scattering of Finite Size Periodic Large Scale System: Simulation by Domain Decomposition Spectral Element Method
*Ma Luo and Qing Huo Liu**

10:50 Paper 4.2. Efficient alternatives to edge vector finite elements for EM field calculations using Hermite interpolation polynomial basis functions on a nodal grid in 2D
J. D. Albrecht, C. R. Boucher, C. I. Ahheng, and L. R. Ram-Mohan

11:10 Paper 4.3. Parallel Finite Element Electromagnetic Code Suite for Accelerator Modeling and Simulation
Cho Ng, Lixin Ge, Kwok Ko, Kihwan Lee, Zenghai Li, Greg Schussman, and Liling Xiao

11:30 Paper 4.4. Optical, Electromagnetics, and Thermal Modeling of Interaction of a Focused Beam of Light with Plasmonic Nanoparticles
*Eren S. Unlu and Kursat Sendur**

11:50 Lunch break

13:10 **Paper 4.5** Finite Element Analysis of Wideband Nanostructures for Photovoltaic Applications
Niccolo Breda, Giuseppe Pelosi, Stefano Selleri, and Ruggero Taddei

13:30 Paper 4.6. Hybrid Optical Waveguides
Ergun Simsek and Qing H. Liu*

Session 6: Advances in Vector Bases for CEM (R. Graglia, A. Peterson, D. Wilton)

13:50 Paper 6.1. On the Construction of Well-conditioned Hierarchical Bases for Tetrahedral H(curl)-Conforming Nedelec Element
Wei Cai and Jianguo Xin*

14:10 Paper 6.2. High Order Finite Elements for Computational Physics: An LLNL Perspective
*Robert N. Rieben**

14:30 Paper 6.3. FEM-MoM-Diakoptic Analysis of Scatterers with Anisotropic Inhomogeneities Using Hierarchical Vector Bases on Large Curved Elements
Ana B. Manić, Dragan I. Olčan, Milan M. Ilić, and Branislav M. Notaros

14:50 Paper 6.4. Rapid, High-Order Finite Element Modelling with FEniCS and SUCEM:FEM
A.J. Otto, E. Lezar, N. Marais, R.G. Marchand, and D.B. Davidson*

15:10 Coffee break

15:40 Paper 6.5. Issues Arising in Connection with Singular Vector Bases
*Roberto D. Graglia and Andrew F. Peterson**

16:00 Paper 6.6. An Orthogonal Prism Vector Bases Based Quadratic Eigenvalue Solver of Linear Complexity for 3-D Electromagnetic Analysis
Jongwon Lee, Duo Chen, and Dan Jiao

16:20 Paper 6.7. Preliminary Research on the Discontinuous Enrichment Method Based Domain Decomposition Scheme for Solving the Three-Dimensional Vector Curl-Curl Equation
Ming-Feng Xue and Jian-Ming Jin*

16:40 Paper 6.8. An Iterative Domain Decomposition Method for Solving Wave Propagation Problems
Gergely Koczka, Kurt Preis, Thomas Bauernfeind, and Oszkár Bíró*

17:00 Paper 6.9. Evaluating Singular and Near-Singular Integrals on Curvilinear Elements
*Donald R. Wilton**

17:20 Technical program ends for the day

Tuesday, June 5, 2012, Billiard Room

Session 7: FETD Modeling of Complex Media and Structures (R. Lee, F. Teixeira)

8:00 Paper 7.1. FETD/FDTD with Automatic Hybrid Mesh Generation
*Shumin Wang**

8:20 Paper 7.2. Study of Numerical Dispersion for Quadrilateral and Triangular Elements Used in the Mixed Finite Element Method
*Luis E. Tobon and Qing H. Liu**

8:40 Paper 7.3. Probabilistic Finite-Difference Time-Domain Method Through Stochastic Electromagnetic Macro-Models
Ata Zadehgo, Andreas C. Cangellaris*

9:00 Paper 7.4. Advancing Ultrahigh Field Human MRI using FDTD Method
*Tamer S. Ibrahim**

9:20 Paper 7.5. The Use of Multiple Time Steps in the FDTD Method
*Ruben Ortega and Robert Lee**

9:40 Coffee break

Session 9: Advances in Hybrid Methods and Multiphysics Problems (B. Shanker, L. Kempel)

10:10 Paper 9.1. Study of the Influence of Thermal Stress on Birefringence of the Microstructured Optical Fibers Using Perturbation Theory
Iury Z. Medeiros, J. A. Mores Jr., Hugo E. H. Figueroa*

- 10:30 Paper 9.2. Thermal-Aware DC IR-Drop Co-Analysis using a Non-conformal Domain Decomposition Approach
Yang Shao, Zhen Peng, Jin-Fa Lee*
- 10:50 Paper 9.3. Development of Time Domain Discontinuous Galerkin-Vector Generalized Finite Element Method
O. Tuncer, B. Shanker, L. C. Kempel*
- 11:10 Paper 9.4. Multi-solver-based Generalized Impedance Boundary Condition for Complicated EM Simulation
Shiquan He, Jun Hu, Zaiping Nie, Lijun Jiang, W. C. Chew*
- 11:30 Paper 9.5. Analysis of Electromechanical Devices Using the Dual-Primal Finite Element Tearing and Interconnecting Method Incorporated with the LU Recombination Method
Wang Yao, Jian-Ming Jin*
- 11:50 Lunch break

- 13:10 Paper 9.6. Alternative Formulations for Hybrid Electromagnetic-Circuit Simulators
*Vivek Subramanian, Ali E. Yilmaz**
- 13:30 Paper 9.7. Symmetric and Nonsymmetric FEM-MoM Techniques Using Higher Order Hierarchical Vector Basis Functions and Curved Parametric Elements
Ana Manić, Milan Ilić, Branislav Notaroš*

Session 11: Domain-Decomposition Methods (J.-F. Lee, Z. Peng)

- 13:50 Paper 11.1. Analysis of Three-Dimensional Array Structures Using Nonconformal and Cement FETI-DP Methods
Ming-Feng Xue and Jian-Ming Jin*
- 14:10 Paper 11.2. A Domain-Decomposition-Based Preconditioner of FE-BI-MLFMA for Computing EM Scattering by Large Inhomogeneous 3D Targets
Ming-Lin Yang, Xin-Qing Sheng, and Cheng-Dan Huang
- 14:30 Paper 11.3. On the Computational Complexity of Domain Wise Multilevel Multifrontal Method
Peng Liu and Chao-Fu Wang*
- 14:50 **Paper 11.4.** Harmonic Balance Domain Decomposition Finite Element for Nonlinear Passive Microwave Devices Analysis
Laurent Ntibarikure, Giuseppe Pelosi, and Stefano Selleri
- 15:10 Coffee break
- 15:40 Paper 11.5. Application of Tree-Cotree Splitting to the Dual-Primal Finite Element Tearing and Interconnecting Method for Solving Low-Frequency Breakdown Problems
Wang Yao and Jian-Ming Jin*
- 16:00 Paper 11.6. Edge Elements and DPM with PBSV to Accelerate Solving Scattering Problems of Electrically Large Objects
Lianyou Sun, Lishan Xue, and Wei Hong*
- 16:20 Paper 11.7. Enhancement of Second-harmonic Generation in an Air-bridge Photonic Crystal Slab: Simulation by Spectral Element Method
*Ma Luo and Qing Huo Liu**
- 16:40 Paper 11.8. A Non-overlapping and Non-conformal Domain Decomposition Method with Optimized Second Order Transmission Conditions for Time-Harmonic Maxwell Equations in R
Zhen Peng and Jin-Fa Lee
- 17:00 Paper 11.9. The Quest for Robust Domain Decomposition Methods in CEM: Are We There Yet?
Marinos N. Vouvakis and Georgios N. Paraschos*
- 17:20 Technical program ends for the day

Tuesday, June 5, 2012, Music Room

Session 8 Adaptive FEM, Higher Order Bases, and Advanced FEM Formulations (J. Webb, M. Gavrilovic, M. Salazar Palma, L. Garcia-Castillo)

- 8:00 Paper 8.1. Hp-adaption for Computing Scattering Parameters on Tetrahedral Meshes
J. P. Webb
- 8:20 Paper 8.2. Efficient RCS Prediction of Jet Engine Air Intakes Using hp-FEM and Reduced Order Modeling
Adam Zdunek and Waldemar Rachowicz*
- 8:40 Paper 8.3. Rules for Adoption of Expansion and Integration Orders in FEM Analysis Using Higher Order Hierarchical Bases on Generalized Hexahedral Elements
Nada J. Šekeljčić, Slobodan V. Savić, Milan M. Ilić, and Branislav M. Notaroš*
- 9:00 Paper 8.4. Auxiliary Space Preconditioning for Hierarchical Elements
A. Aghabarat, J. P. Webb
- 9:20 Paper 8.5. Chain Homotopy Operators and Whitney form-based Finite Element Analysis
*P. Robert Kotiuga**
- 9:40 Coffee break
- 10:10 Paper 8.6. Comparative Performance of Nodal-Based Versus Edge-Based Finite Element Formulations
Ruben Otin, Santiago Badia, and Luis E. Garcia-Castillo*
- 10:30 Paper 8.7. Discontinuous Petrov-Galerkin Methods for Wave Propagation with Negligible Pollution Errors
Jay Gopalakrishnan, Leszek Demkowicz, and Igancio Muga*
- 10:50 Paper 8.8. A Three-Dimensional Self-Adaptive hp Finite Element Method for the Characterization of Waveguide Discontinuities
Ignacio Gomez-Revuelto, Luis E. Garcia-Castillo, Sergio Llorente-Romano, and David Pardo

Session 10: FEM Modeling and Applications of Metamaterials and Periodic Media (K. Sertel, J. Volakis)

- 11:10 Paper 10.1. Finite Element Modeling of Magnetic Metamaterials and Associated Challenges for Their Experimental Characterization

11:10 Paper 10.1. Finite Element Modeling of Magnetic Metamaterials and Associated Challenges for their Experimental Characterization
Nil Apaydin, Kubilay Sertel, and John L. Volakis*

11:30 Paper 10.2. A Method for Rapid Evaluation of Space-time Convolutions with the Time Domain Periodic Green's Function
*D. Dault and B. Shanker**

11:50 Lunch break

13:10 Paper 10.3. Finite Element Analysis of Photon Density of States for Two-Dimensional Photonic Crystals with Omnidirectional Light Propagation
R. F. Jao and M. C. Lin

13:30 Paper 10.4. A Hybrid Approach for Characterizing Large Anisotropic and Inhomogeneous Metamaterial Structures
*Justin G. Pollock and Ashwin K. Iyer**

Session 12: Advanced FEM/MoM Modeling, Design, and Optimization (C.J. Reddy, U. Jakobus, J. Zapata, J. Gil, D. Jiao, A. Cangelaris)

13:50 Paper12.1. Overview of the Hybrid Finite Element Method/Method of Moments Capability in FEKO
Marianne Bingle, Ulrich Jakobus, and Johann J. van Tonder*

14:10 Paper12.2. Application of Hybrid FEM/MoM Technique for Handheld Mobile Devices
Rensheng Sun and C. J. Reddy

14:30 Paper12.3. Application of Hybrid FEM/MoM Technique for Cavity Backed Apertures
Gopinath Gampala, Rensheng Sun and C. J. Reddy

14:50 Paper12.4. Eigenproblem Approach for Analysis and Optimization of Microwave Filters Using Finite Element Method
Adam Lamecki, Michal Mrozowski

15:10 Coffee break

15:40 Paper12.5. Does Sensitivity Analysis for Fast Frequency Sweeps Converge?
*Andreas Köhler, Ortwin Farle, Romanus Dyczij-Edlinger**

16:00 Paper12.6. Benchmark of Conformal Finite-Difference Method against Finite Element Method
M. C. Lin

16:20 Paper12.7. Layout-Integrated Electromagnetic Interconnect Characterization and Simulation
Dennis Nagle, Jilin Tan, Jian-Ming Jin

16:40 Paper12.8. An Extraction-Free Circuit Simulator of Linear Complexity Guided by Electromagnetics-Based First Principles
Qing He, Duo Chen, and Dan Jiao

17:00 Paper12.9. A model reduction method for multiple particle electromagnetic scattering in three dimensions
M. Ganesh, J. Hesthaven, and B. Stamm*

17:20 Technical program ends for the day

Wednesday, June 6, 2012

15:00 **Open forum discussion 1: Open Forum Discussion on FEM and CEM Bases, Elements, and Formulations**
 (A. Peterson, D. Wilton, J. Webb, K. Sertel)

15:45 **Open forum discussion 2: Open Forum Discussion on DDM, MOR, and Efficient FEM and Hybrid Solutions**
 (J.-F. Lee, Zhen Peng, R. Dyczij-Edlinger, V. de la Rubia)

16:30 Coffee break

16:50 **Open forum discussion 3: Open Forum Discussion on FEM-Based Modeling, Design, and Applications**
 (L. Kempel, B. Shanker, R. Kindt, C.J. Reddy, K. Zhao, T. Euler)

18:00 Technical program ends for the day